

Solutions: Additional Exercises 2

General Series

Series and Integral Transforms

Exercise 1 — Telescoping Sums (Exact Values)

(a) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$

Partial fractions: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. Telescoping:

$$S_N = 1 - \frac{1}{N+1} \rightarrow \boxed{1}.$$

(b) ★ $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$

Partial fractions: $\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right)$.

This is a *step-2 telescope*. Grouping:

$$S_N = \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2} \right) \rightarrow \frac{1}{2} \cdot \frac{3}{2} = \boxed{\frac{3}{4}}.$$

(c) ★★ $\sum_{n=2}^{\infty} \ln \left(1 - \frac{1}{n^2} \right)$

Factor: $1 - \frac{1}{n^2} = \frac{n^2 - 1}{n^2} = \frac{(n-1)(n+1)}{n^2}$. Using log rules:

$$\ln \left(\frac{(n-1)(n+1)}{n^2} \right) = \underbrace{(\ln(n-1) - \ln n)}_{u_n} + \underbrace{(\ln(n+1) - \ln n)}_{v_n}.$$

For the partial sum S_N :

$$\sum_{n=2}^N u_n = (\ln 1 - \ln 2) + (\ln 2 - \ln 3) + \cdots + (\ln(N-1) - \ln N) = \ln 1 - \ln N = -\ln N,$$

$$\sum_{n=2}^N v_n = (\ln 3 - \ln 2) + (\ln 4 - \ln 3) + \cdots + (\ln(N+1) - \ln N) = \ln(N+1) - \ln 2.$$

Hence $S_N = -\ln N + \ln(N+1) - \ln 2 = \ln \left(\frac{N+1}{N} \right) - \ln 2 \rightarrow 0 - \ln 2 = \boxed{-\ln 2}.$

(d) ★★ $\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)}$

Use the identity $\frac{1}{n(n+1)(n+2)} = \frac{1}{2} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right)$.

Telescoping:

$$S_N = \frac{1}{2} \left(\frac{1}{1 \cdot 2} - \frac{1}{(N+1)(N+2)} \right) \rightarrow \frac{1}{2} \cdot \frac{1}{2} = \boxed{\frac{1}{4}}.$$

Exercise 2 — Alternating Series: Classification

For each series we determine whether it **diverges**, is **conditionally** convergent, or **absolutely** convergent.

(a) ★ $\sum \frac{(-1)^{n+1}}{n^2 + 1}$

$|a_n| = \frac{1}{n^2 + 1} \leq \frac{1}{n^2}$ and $\sum 1/n^2$ converges ($p = 2$). By Comparison, $\sum |a_n|$ converges.
Absolutely convergent.

(b) ★ $\sum \frac{(-1)^n}{\sqrt{n}}$

$\sum 1/\sqrt{n}$ diverges ($p = 1/2$), so not absolutely convergent. However, $1/\sqrt{n} \searrow 0$, so **Leibniz's Test** applies.
Conditionally convergent.

(c) ★ $\sum (-1)^n \frac{n+1}{n}$

$a_n = \frac{n+1}{n} \rightarrow 1 \neq 0$. By the **Necessary Condition** the series **diverges**.

(d) ★★ $\sum (-1)^n \frac{\ln n}{n}, n \geq 2$

$\sum \frac{\ln n}{n}$: since $\ln n \geq 1$ for $n \geq 3$, we have $\frac{\ln n}{n} \geq \frac{1}{n}$, so by Comparison with the harmonic series, $\sum |a_n|$ **diverges**.

For conditional convergence: $f(x) = \ln x/x$ satisfies $f'(x) = (1 - \ln x)/x^2 < 0$ for $x > e$, so the sequence is eventually decreasing, and $(\ln n)/n \rightarrow 0$. By **Leibniz's Test**, the series **converges conditionally**.

$$(e) \star \sum \frac{(-1)^n}{n(n+1)}$$

$\frac{1}{n(n+1)} \sim \frac{1}{n^2}$; **Limit Comparison** with $1/n^2$ gives $L = 1$, so $\sum |a_n|$ converges. **Absolutely convergent.**

$$(f) \star\star \sum (-1)^n \left(1 - \cos \frac{1}{n}\right)$$

For small u , $1 - \cos u \sim u^2/2$, so $1 - \cos(1/n) \sim 1/(2n^2)$. **Limit Comparison:** $\lim n^2(1 - \cos(1/n)) = 1/2$. Since $\sum 1/n^2$ converges, $\sum |a_n|$ converges. **Absolutely convergent.**

$$(g) \star\star \sum (-1)^n \left(\sqrt{n+1} - \sqrt{n}\right)$$

Rationalise: $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \sim \frac{1}{2\sqrt{n}}$.

Absolute series: $\sum \frac{1}{\sqrt{n+1} + \sqrt{n}} \sim \sum \frac{1}{2\sqrt{n}}$ diverges.

The terms decrease to 0, so **Leibniz's Test** applies. **Conditionally convergent.**

$$(h) \star\star \sum (-1)^n \left(1 - \left(\frac{n}{n+1}\right)^n\right)^{1/2}$$

Now $\left(\frac{n}{n+1}\right)^n = \left(1 - \frac{1}{n+1}\right)^n \rightarrow e^{-1}$, so:

$$1 - \left(\frac{n}{n+1}\right)^n \rightarrow 1 - \frac{1}{e} \neq 0 \implies a_n \rightarrow \sqrt{1 - 1/e} \neq 0.$$

Diverges by the Necessary Condition.

$$(i) \star\star\star \sum (-1)^n \frac{(2n)!}{4^n (n!)^2}$$

By Stirling: $\frac{(2n)!}{4^n (n!)^2} \sim \frac{1}{\sqrt{\pi n}} \rightarrow 0$, and this is decreasing (ratio $(2n+1)/(2n+2) < 1$).

Leibniz's Test applies $\implies$ converges. But $\sum \frac{1}{\sqrt{\pi n}}$ diverges ($p = 1/2$).

Conditionally convergent.

Exercise 3 — General Convergence

$$(a) \star \sum \frac{\cos(2\pi n)}{n^2}$$

Note $\cos(2\pi n) = 1$ for all $n \in \mathbb{N}$. So the series equals $\sum 1/n^2$, which converges ($p = 2$). **Absolutely convergent.**

(b) ★ $\sum \frac{n}{e^n}$

Ratio Test: $\frac{(n+1)/e^{n+1}}{n/e^n} = \frac{n+1}{en} \rightarrow \frac{1}{e} < 1$. **Converges absolutely.**

(c) ★ $\sum \frac{1}{n^{1+a}}, a > 0$

This is a p -series with $p = 1 + a$. Converges $\iff p > 1 \iff a > 0$. **Converges for all $a > 0$.**

(d) ★★ $\sum \frac{1}{n \sin(1/n)}$

For small u , $\sin u \sim u$, so $n \sin(1/n) \rightarrow 1$ and $\frac{1}{n \sin(1/n)} \rightarrow 1 \neq 0$. By the **Necessary Condition**, the series **diverges**.

(e) ★★ $\sum \frac{(1 + 1/n)^{n^3}}{n^3}$

$(1 + 1/n)^n \rightarrow e$, so $(1 + 1/n)^{n^3} = [(1 + 1/n)^n]^{n^2} \sim e^{n^2} \rightarrow +\infty$. Hence $a_n \rightarrow +\infty \neq 0$. **Diverges.**

(f) ★★★ $\sum \frac{(2n)!}{4^n (n!)^2}$

As in Exercise 2(i), by Stirling: $a_n \sim 1/\sqrt{\pi n}$. **Limit Comparison** with $1/\sqrt{n}$ (which diverges, $p = 1/2$) gives $L = 1/\sqrt{\pi} \neq 0, \infty$. **Diverges.**

(g) ★★★ $\sum \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots (2n)}$

Ratio Test: $\frac{a_{n+1}}{a_n} = \frac{2n+1}{2n+2} \rightarrow 1$. Inconclusive.

By Wallis's product, $a_n = \frac{(2n-1)!!}{(2n)!!} \sim \frac{1}{\sqrt{\pi n}}$. Since $\sum 1/\sqrt{\pi n}$ diverges, the series **diverges**.

(h) ★★★ $\sum \frac{(2n-1)!!}{(2n)!!} \cdot 3^{n/3}$

Using $a_n \sim 1/\sqrt{\pi n}$ and multiplying by $3^{n/3}$, the terms grow like $3^{n/3}/\sqrt{\pi n} \rightarrow +\infty$. **Diverges** by the Necessary Condition.

Exercise 4 — Theoretical Arguments

(a) ★★ $\sum a_n$ converges $\Rightarrow \sum a_n/n^2$ converges absolutely

Since $\sum a_n$ converges, $a_n \rightarrow 0$, so $|a_n| < 1$ for all sufficiently large n . Therefore for large n :

$$\left| \frac{a_n}{n^2} \right| = \frac{|a_n|}{n^2} \leq \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges, by the **Comparison Test**, $\sum |a_n/n^2|$ converges. $\square$

(b) ★★★ $\sum a_n$ converges $\Rightarrow \sum \sin(a_n/n^3)$ converges absolutely

Since $\sum a_n$ converges, $a_n \rightarrow 0$, so for large n , $|a_n/n^3| \leq |a_n|/n^3 \leq |a_n|$ which is eventually < 1 , i.e. the argument of sine is small. Using $|\sin u| \leq |u|$:

$$\left| \sin\left(\frac{a_n}{n^3}\right) \right| \leq \frac{|a_n|}{n^3} \leq \frac{|a_n|}{1} \cdot \frac{1}{n^3}.$$

More carefully: for large n , $|a_n| \leq 1$, so $|a_n/n^3| \leq 1/n^3 \leq 1$, giving $|\sin(a_n/n^3)| \leq |a_n|/n^3 \leq |a_n|$. Since $\sum |a_n|$ converges (actually we only know $\sum a_n$ converges), apply instead: $|\sin(a_n/n^3)| \leq 1/n^3$, and $\sum 1/n^3$ converges. **Converges absolutely.** $\square$

(c) ★★ **Proof of Leibniz's Test**

Let $a_n \searrow 0$. Define $S_N = \sum_{n=1}^N (-1)^{n+1} a_n$.

Even partial sums: $S_{2N} = S_{2N-2} + (a_{2N-1} - a_{2N}) \geq S_{2N-2}$ (non-decreasing), and $S_{2N} = a_1 - (a_2 - a_3) - \cdots - (a_{2N-2} - a_{2N-1}) - a_{2N} \leq a_1$ (bounded above).

Odd partial sums: $S_{2N+1} = S_{2N} + a_{2N+1}$ where $a_{2N+1} \rightarrow 0$, so $S_{2N+1} \rightarrow L$.

Both subsequences converge to the same limit L , so $S_N \rightarrow L$.

Error estimate: $|S - S_N| \leq a_{N+1}$. $\square$

(d) ★★ **Approximating $\sum \frac{(-1)^{n+1}}{n(n+1)^2}$ to within 10^{-5}**

This is an alternating series with $a_n = 1/(n(n+1)^2)$, decreasing to 0. By the **Leibniz error bound**, we need the first omitted term:

$$a_{N+1} = \frac{1}{(N+1)(N+2)^2} < 10^{-5}.$$

$$\text{Testing } N = 46: a_{47} = \frac{1}{47 \cdot 48^2} = \frac{1}{108288} \approx 9.2 \times 10^{-6} < 10^{-5}.$$

$$\text{Testing } N = 45: a_{46} = \frac{1}{46 \cdot 47^2} = \frac{1}{101614} \approx 9.8 \times 10^{-6} < 10^{-5}.$$

So summing the first **N = 45** terms gives error $< 10^{-5}$.

Key takeaways: Alternating series are classified by checking absolute convergence first (Comparison/ p -test), then conditional convergence via Leibniz. The necessary condition eliminates many divergent cases quickly. Stirling's approximation is essential for double-factorial and binomial-coefficient series.